

3. Change a character emotion (Neutral to happy, Happy to sad, sad to angry etc.,) using Unity.

Step 1: Prepare Emotion Images

You need face images like:

- 😐 Neutral
- 😊 Happy
- 😞 Sad
- 😡 Angry

📁 Save them as PNG files:

```
neutral.png  
happy.png  
sad.png  
angry.png
```

Step 2: Import into Unity

1. Drag images into **Assets folder**
2. Select each image:
 - Texture Type → **Sprite (2D and UI)** ✓
 - Apply

OPTION B: 3D Character (Plane Method)

If your scene is 3D, do this instead:

✓ Step 3.1: Create Plane

Right-click → 3D Object → Quad

Rename:

CharacterFace

✓ Step 3.2: Create Material

1. Right-click → Create → Material
 2. Name → `NeutralMat`
-

✓ Step 3.3: Assign Texture

1. Select `NeutralMat`
 2. Drag:
📁 `neutral.png` into **Base Map**
-

✓ Step 3.4: Apply Material

- Drag `NeutralMat` onto the Quad
-

✓ Step 3.5: Rotate Properly

Set rotation:

X: 0
Y: 180 (or 0 depending on camera)
Z: 0

🎯 What you have now:

📁 A flat 3D face visible in scene

Step 4: Create Materials (for each emotion)

1. Right-click → Create → Material
2. Create:
 - `NeutralMat`
 - `HappyMat`
 - `SadMat`
 - `AngryMat`
3. For each material:
 - Drag corresponding texture into **Base Map**

Step 5: Create Script

Create script → `EmotionController`

```

using System.Diagnostics;
using UnityEngine;
using UnityEngine.InputSystem;

public class EmotionController : MonoBehaviour
{
    public Renderer characterRenderer;

    public Material neutralMat;
    public Material happyMat;
    public Material sadMat;
    public Material angryMat;

    int currentEmotion = 0;

    void Update()
    {
        if (Mouse.current.leftButton.wasPressedThisFrame)
        {
            UnityEngine.Debug.Log("Clicked"); // ➡ ADD THIS
            ChangeEmotion();
        }
    }

    void ChangeEmotion()
    {
        currentEmotion = (currentEmotion + 1) % 4;

        switch (currentEmotion)
        {
            case 0:
                characterRenderer.material = neutralMat;
                break;
            case 1:
                characterRenderer.material = happyMat;
                break;
            case 2:
                characterRenderer.material = sadMat;
                break;
            case 3:
                characterRenderer.material = angryMat;
                break;
        }
    }
}

```

STEP 6: Attach Script + Assign Everything

This step has **2 parts**:

❑ PART A: Attach Script to Character

1. Go to **Hierarchy**
2. Click your object:
 ☞ Character (Sprite or Quad you created)
3. Go to **Project window**

4. Find your script:

🔍 EmotionController

5. **Drag the script onto the Character**

🔍 OR:

- Select Character
- In Inspector → Click **Add Component**
- Search: EmotionController

✓ Now script is attached

📺 PART B: Assign Variables in Inspector

Now select your **Character** again

In Inspector you'll see:

Emotion Controller (Script)

Inside it:

- Character Renderer
 - Neutral Mat
 - Happy Mat
 - Sad Mat
 - Angry Mat
-

👁️ **Now assign each one:**

1 Character Renderer

🔍 This tells Unity **which object to change**

- Drag your **Character object** into:

Character Renderer

✓ If using 3D (Quad) → works perfectly

✓ If using Sprite → see note below

Assign Materials

From Project window, drag:

Field	What to drag
Neutral Mat	NeutralMat
Happy Mat	HappyMat
Sad Mat	SadMat
Angry Mat	AngryMat

 Drag one by one into correct slots